M5: Final Capstone Full Report

Chapter 5

Jake Gapusan

CIS 480

Professor Kashihara

I. Summary of Hypothesis and Results

The following project developed four hypotheses centered on AI adoption, employment patterns, skills, productivity, and inequality. Each hypothesis was evaluated using labor market datasets, the Job Threat Index, ChatGPT adoption metrics, socioeconomic indicators, and public opinion data.

Hypothesis # 1

- ***Null Hypothesis*** (H_{01}, two-tailed) - No relationship between AI adoption and employment growth.
- ***Alternative Hypothesis*** (H_{11}, one-tailed) - Higher AI adoption is associated with lower employment growth in routine industries.

Result: Not rejected (inconclusive).

Our Job Threat Index dataset showed which occupations are vulnerable, but we did not have industry-level employment growth data to statistically confirm displacement. While many routine jobs showed high automation risk, we could not test growth directly. Thus, results remain inconclusive; not enough evidence to reject the null.

Hypothesis # 2

- ***Null Hypothesis*** (H_{01}, two-tailed) - AI adoption does not affect demand for digital or analytical skills.
- ***Alternative Hypothesis*** (H_{11}, one-tailed) - Higher AI adoption increases demand for digital and analytical skills.

Result: Rejected

Public opinion data, user reviews, and ChatGPT usage trends show high adoption for tasks requiring writing, analysis, coding support, and decision-making. Job Threat Index patterns also show that non-routine, analytical, and technical jobs are the least threatened, implying rising skill demand. Evidence aligns strongly with the literature (OECD, Stanford HAI), so the null hypothesis is rejected.

Hypothesis # 3

- ***Null Hypothesis*** (H_{01}, two-tailed) - AI-driven productivity gains are unrelated to worker experience
- ***Alternative Hypothesis*** (H_{11}, one-tailed) - AI benefits less-experienced workers more

Result: Rejected

Our public opinion and review sentiment show that new or casual ChatGPT users report major productivity boosts, especially for learning, writing help, and coding guidance. This mirrors Brynjolfsson et al. (2023), where novice employees benefited most. Despite not having direct employee-level data, the behavioral evidence strongly supports the alternative hypothesis.

Hypothesis # 4

- ***Null Hypothesis*** (H_{01}, two-tailed) - AI adoption has no association with wage inequality
- ***Alternative Hypothesis*** (H_{11}, one-tailed) - AI adoption increases wage polarization.

Result: Not rejected

County socioeconomic data (income, unemployment, education) shows inequality patterns, but AI adoption itself cannot be isolated as the cause. We cannot confirm wage polarization statistically, although the Job Threat Index suggests uneven impacts across occupations. Therefore, the evidence is not strong enough to reject the null.

II. Project Summary and Reflection

The purpose of this project is to investigate whether AI automation has already begun reshaping society and how it impacts everything, from the jobs we work to the skills we need to the way we use AI tools like ChatGPT in our personal lives. Using multiple datasets, industry literature, and visual analytics through Python and Power BI, the project analyzed AI adoption from three major angles: job automation risk, public usage behavior, and socioeconomic conditions across U.S. counties.

Several key findings emerged. First, AI adoption is accelerating rapidly. ChatGPT usage alone showed massive monthly growth, confirming that AI tools are becoming mainstream in both personal and professional contexts. Second, the Job Threat Index revealed that routine and clerical jobs face the highest automation risk, while analytical and technical roles remain the most resilient. Third, socioeconomic indicators show that education and income levels strongly correlate with AI vulnerability; counties with lower bachelor's degree rates and lower incomes face higher unemployment and a higher concentration of vulnerable jobs.

Furthermore, the project also came with challenges. Cleaning datasets from multiple sources required extensive formatting, merging, and error-checking. Many datasets lacked direct variables needed for statistical testing, requiring careful interpretation rather than running formal regression models. Power BI design also required ensuring mobile accessibility and building clear, intuitive visuals.

Overall, this project strengthened my skills in data cleaning, modeling, visualization, and interpretation. It also deepened my understanding of how societal-scale data can reflect real-world trends. Moving forward, I plan to expand this work with additional datasets, particularly long-term employment data and wage distribution measures, to more rigorously evaluate the hypotheses.

III. Real-World Implications

Introduction to Real-World Impacts

AI adoption is no longer theoretical. It is reshaping nearly every sector of society, and the datasets analyzed in this project provide clear evidence of how these shifts are emerging. The findings demonstrate that AI tools like ChatGPT are not only experiencing explosive growth but are actively influencing how individuals learn, work, and interact with information. Meanwhile, job-level automation risk highlights which workers may face displacement, and socioeconomic mapping reveals how vulnerable populations may be disproportionately affected. These insights have major implications for the future of healthcare, education, transportation, workforce development, income inequality, and public policy.

1) The Bigger Picture of AI's Impact

Working on this project made me realize how deep AI's influence already is, even if most people don't think about it. At first, AI feels like something far away, something only big tech companies worry about. However, upon analyzing actual data on ChatGPT usage, job automation risk, and county-level socioeconomic conditions, I discovered that AI is already shaping everyday life in ways we often overlook. It affects how people work, learn, ask questions, solve problems, and even how they make decisions.

Moreover, the rapid growth of ChatGPT alone demonstrates that millions of people use AI every month to accomplish something meaningful for themselves, whether that's writing, studying, coding, or addressing life's problems. And when I compared that with the Job Threat Index, it became clear that the jobs most at risk are the ones people perform every day in offices, retail stores, call centers, and administrative positions. This means AI is not just a futuristic idea; it's something that can directly impact worker security, job stability, and the kind of skills people need to stay competitive.

Furthermore, the socioeconomic data gave another important layer. Counties with lower education levels and lower incomes also showed higher unemployment rates. These same areas often have more jobs considered "high-risk" for AI replacement. This suggests that AI could repeat patterns we've seen in other technological revolutions: those who already have fewer opportunities might be hit the hardest. That's why understanding these trends matters for public policy, education systems, and employers who will need to support workers as technology continues to evolve.

2) Implications for Education

One of the clearest findings from this project is that AI is completely changing how people learn. The public opinion and reviews datasets showed that a lot of users depend on ChatGPT to help with school assignments, explanations, and new skills. Students use AI for writing support, research summaries, problem-solving, and even coding help. This means education can no longer ignore AI because it's already here, and students are using it whether schools approve or not.

The implication is that teachers and schools need to shift from preventing AI usage to teaching students how to use it properly. Just like calculators didn't destroy math, AI won't destroy learning, but there needs to be guidance. Students should learn the difference between relying on AI and using it as a tool. They should learn how to check AI's accuracy, how to integrate it into their assignments, and how to use it to increase productivity rather than replace thinking.

In addition, the data also shows that AI tools especially help beginners or learners who need extra support. This lines up with the rejected null hypothesis, which AI benefits people who are not experts yet. In a classroom setting, that could mean fewer students feeling left behind. AI could reduce learning gaps if schools adapt and train students in ethical, informed usage.

However, the risk is also clear: if wealthier districts have the resources to teach AI skills while underserved districts don't, the education gap will widen even more. This is where public policy needs to step in so that all students, regardless of background, gain exposure to modern tools.

3) Implications for Workforce & Income Inequality

The Job Threat Index gave the most direct picture of how AI might reshape the workforce. Jobs that involve predictable routines, such as clerical work, customer service, bookkeeping, scheduling, and administrative tasks, carry the highest automation risk. These are not niche positions; these are jobs millions of people depend on.

Moreover, this connects directly to income inequality. When I mapped the county-level socioeconomic data, it was clear that counties with lower degrees and lower earnings also had higher unemployment rates. These counties are also the ones filled with jobs that fall into "high automation risk" categories. The implication is that AI could make inequality worse if society doesn't prepare. Workers in vulnerable areas need retraining programs, new career pathways, and access to AI-related education.

Even though Hypothesis 4 remained inconclusive, the data still hinted at a relationship between AI and economic disparities. If AI replaces jobs faster in low-income areas, those communities might face unemployment spikes and wage stagnation. Meanwhile, high-education regions where jobs require creativity, analysis, and technical skills may benefit from AI rather than be harmed by it.

This is why workforce development programs matter. People need training in digital skills, data literacy, and analytical thinking. The good news is that AI also creates new job opportunities, such as prompt engineers, automation managers, AI support specialists, data analysts, ML operators, and more. But we need pathways to help people transition into these roles. Without proper support, AI could widen the gap between high-skill and low-skill workers.

4) Implications for Public Policy and Society

From a public policy perspective, the results show several urgent needs:

a. **Education Reform**:

Schools need to update curriculum to include responsible AI usage, digital literacy, and new skill development.

b. **Job Transition Support**:

Governments should invest in upskilling programs for workers in routine jobs. If these workers are not supported, automation could cause sudden job loss in certain sectors.

c. **Monitoring Inequality**:

AI adoption must be studied within the context of income and unemployment rates across counties. Without monitoring, policymakers might overlook which communities are being left behind.

d. **Ethical AI Guidelines**:

As more people rely on AI for decision-making, there needs to be protection for privacy, misinformation, and bias. Public opinion data showed that while many trust AI, many are also afraid of AI making major decisions. This fear is reasonable and should encourage responsible AI deployment.

e. **Transparency in AI Usage**:

Organizations should be open about when they use AI for hiring, firing, scheduling, and performance evaluations. Without transparency, workers may feel powerless or misjudged by algorithms they don't understand.

Overall, AI's impact is not limited to businesses or tech companies. Every year, more people rely on AI to help with daily tasks, from writing emails to navigating healthcare information. The more integrated AI becomes, the more important it is to guide its use and ensure fairness.

5) *Implications for Healthcare, Transportation, and Daily Life*

AI tools are already influencing healthcare by helping people research symptoms, prepare questions before appointments, and understand medical information better. This empowers patients, especially those from underserved communities who may not always have access to high-quality medical guidance. However, the risk is that AI also spreads misinformation when not monitored correctly. Healthcare providers must remain the main authority.

In transportation, automation and self-driving technology continue to grow. While our dataset didn’t focus on transportation directly, the same principles apply: routine driving jobs could be at risk long-term, requiring new labor policies and retraining.

In daily life, AI has become a personal assistant for millions. People rely on it to plan, write, think, organize, and learn. This changes how people solve problems; they no longer do everything alone. Instead, they collaborate with AI almost like a partner. This can increase productivity and creativity, but it also requires people to maintain critical thinking skills and avoid over-dependence.

Thus, the biggest implication is that AI is not “coming soon” because it is already here. Workers, students, policymakers, and families must adapt to a world where AI is woven into normal routines. This project helped me understand that AI automation isn’t just a technological challenge; it’s a social one. How society responds will determine whether AI becomes a source of progress or a source of inequality.

Sources

Stanford Institute for Human-Centered Artificial Intelligence. (2025). AI Index Report 2025. https://hai.stanford.edu/ai-index/2025-ai-index-report

AIPRM. (2024). AI Statistics: 2024 Trends & Insights. https://www.aiprm.com/ai-statistics/

AIPRM. (2024). How Many People Use AI in 2024? https://www.aiprm.com/ai-statistics/#how-many-people-use-ai

unclepablo. (2024). AI Impact on Jobs [Notebook]. Kaggle. https://www.kaggle.com/code/unclepablo/ai-impact-on-jobs

Exploding Topics. (2024). ChatGPT Users Statistics (2024). https://explodingtopics.com/blog/chatgpt-users

Funk, C., & Kennedy, B. (2023, February 15). Public Awareness of Artificial Intelligence in Everyday Activities. Pew Research Center. https://www.pewresearch.org/science/2023/02/15/public-awareness-of-artificial-intelligence-in-everyday-activities/ps_2023-02-15_ai-awareness_00-01-png/

DemandSage. (2024). ChatGPT Statistics and User Growth (2024). https://www.demandsage.com/chatgpt-statistics/

Business of Apps. (2024). ChatGPT Statistics (2024). https://www.businessofapps.com/data/chatgpt-statistics/

Manavgupta92. (2023). From Data Entry to CEO: The AI Job Threat Index [Dataset]. Kaggle. https://www.kaggle.com/datasets/manavgupta92/from-data-entry-to-ceo-the-ai-job-threat-index

Stanford Institute for Human-Centered Artificial Intelligence. (2025). AI Index 2025 – Supplementary Materials [Google Drive folder]. https://drive.google.com/drive/folders/1AxxxL9-AsaeMdDKtTNHCR1KqEJTsHCod

Annual unemployment rate by state (2018–2023). PolicyMap, https://www-policymap-com.ezp.mesacc.edu/newmaps/e/mesacc (based on data from Bureau of Labor Statistics; accessed September 5, 2025).